

Threshold-Based Unknown Species Detection — Full Experiment Report

Experiment ID: `threshold-full-001`

June 2026

Experiment Category: Autoresearch — Retrieval Threshold Optimization

Repository: MycoAI Monorepo — `research/src/experiments/threshold/`

Test Data: 471 known (7 species, 15 test strains) + 1,173 unknown (41 species, 50 strains)

Formulas Evaluated: 162 unique formulas $\times$ 3 algorithms = 528 experiments

Key Finding

Best result: `gap01_sq + f1_grid` achieves $F1 = 0.4461$ (Precision = 0.2876, Recall = 0.9936, Specificity = 0.0119).

Severe class imbalance dominates: the model labels nearly everything as “known,” yielding high recall but near-zero specificity. F1 alone is misleading for this task.

1 Executive Summary

1.1 Objective

This experiment evaluates whether a score-based thresholding pipeline can reliably separate **known** fungal species (present in the training index) from **unknown** species (novel to the system). For each query image, a KNN retrieval returns the top- K neighbours with similarity scores $(s_0, s_1, \dots, s_4)$. A *formula* maps these five scores to a scalar confidence value; a threshold applied to this value classifies the image as known ($\geq t$) or unknown ($< t$). The goal is to find the formula–algorithm–threshold combination that maximises the F1 score for the known class.

1.2 Configuration

The retrieval pipeline uses an EfficientNetB1 model fine-tuned on fungal colony features, querying the `myco_fungi_features_full_finetuned` Qdrant collection with $K = 11$ neighbours under environment strategy E1 (same growth medium only) and weighted score aggregation.

1.3 Test Data

The test set comprises **471 known samples** drawn from 7 *Penicillium* species held out from the Qdrant index during retrieval (15 test strains across 7 growth environments) and **1,173 unknown samples** from 41 diverse fungal species (50 strains) never seen during training.

1.4 Best Result

The `gap01_sq` formula (squared difference between top-1 and top-2 scores) paired with the `f1_grid` threshold-finding algorithm yields $F1 = 0.4461$ at threshold $t = 4.62 \times 10^{-7}$. This configuration correctly accepts 468 of 471 known samples (recall = 99.4%) but incorrectly accepts 1,159 of 1,173 unknown samples as known (specificity = 1.2%).

1.5 Limitation

F1 proves misleading as a primary metric for this task. The severe class imbalance (1,173 unknowns vs. 471 knowns) means that a strategy maximising F1 under standard grid search will always favour near-universal acceptance — because recall dominates the F1 numerator when the threshold is set near zero. The result is a classifier with >99% recall but effectively no ability to reject unknowns. Future work must incorporate false-positive-rate constraints, weighted loss functions, or a two-stage classification architecture.

2 Experiment Setup

2.1 Retrieval Pipeline

The retrieval configuration is fixed across all experiments:

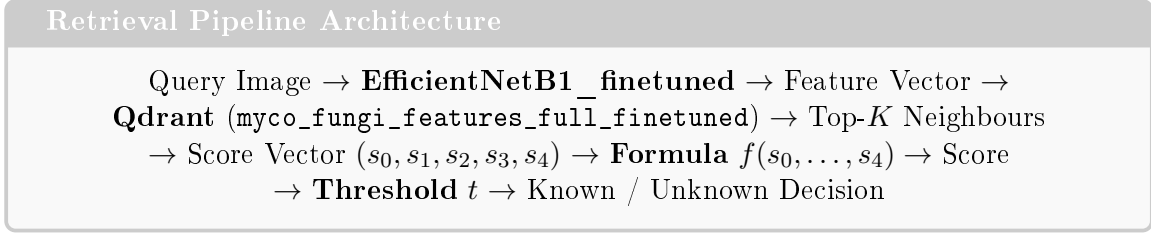


Figure 1: End-to-end retrieval and thresholding pipeline. The formula maps neighbour scores to a scalar; the threshold determines the known/unknown classification.

Fixed Retrieval Parameters		
Parameter	Value	Description
Collection	myco_fungi_features_full_finetuned	Qdrant vector store
Extractor	EfficientNetB1_finetuned	Feature extractor backbone
K	11	Top neighbours retrieved
Environment	E1	Same growth medium only
Aggregation	Weighted	Score-weighted neighbour voting

2.2 Test/Train Split

The Qdrant index contains embeddings for 53 known *Penicillium* species (the curated dataset). Seven species were **held out** from the index during retrieval to serve as *known* test samples, while the remaining 46 species constitute the training corpus. Additionally, 41 *unknown* species from the incoming (diverse) dataset — species entirely absent from the curated collection — were retrieved against the index.

2.3 Dataset Statistics

Table 1: Dataset composition for threshold experiment. Known samples span 7 species with 15 strains tested across 7 growth environments. Unknown samples span 41 species across 50 strains and 8 environments.

Statistic	Known	Unknown	Total
Samples	471	1,173	1,644
Unique species	7	41	48
Unique strains	15	50	65
Growth environments	7	8	8
Images per species (mean)	67.3	28.6	34.3

The known species and their sample counts are: *P. freii* (102), *P. polonicum* (99), *P. melanoconidium* (54), *P. viridicatum* (42), *P. tricolor* (78), *P. neoehinulatum* (54), and *P. aurantiogriseum* (42). The growth environments span CYA (84), MEA (84), CREA (78), YES (72), DG18 (69), CYA30 (42), and CYAS (42).

The unknown dataset is more diverse but shallower per species. The five most frequent unknown species are *P. nalgiovense* (90 samples), *E. crustaceum* (69), *P. expansum* (63), *P. griseofulvum* (60), and *E.gyptiacum* (36). Among environments, CYA and MEA dominate (276 and 270 samples respectively).

2.4 Formula Design Space

A **formula** is a function $f(s_0, s_1, s_2, s_3, s_4) \rightarrow \mathbb{R}$ that maps the five neighbour similarity scores to a single scalar confidence value. The design space explored 162 distinct formulas grouped into qualitative families:

Table 2: Formula families and their naming conventions. Each family represents a qualitatively distinct mathematical transformation of the neighbour score vector.

Prefix	Count	Description
gap_{i}_{j}	10	Score differences $s_i - s_j$
gnorm_{i}_{j}	10	Normalised gaps $(s_i - s_j)/(s_i + s_j)$
ratio_{i}_{j}	20	Score ratios s_i/s_j
prod_{i}_{j}	10	Score products $s_i \times s_j$
avg_top{k}	4	Arithmetic mean of top- k
gm_top{k}	4	Geometric mean of top- k
hm_top{k}	4	Harmonic mean of top- k
ne_top{k}	4	Normalised entropy of top- k
std_top{k}	4	Standard deviation of top- k
range_top{k}	4	Range of top- k
exp_decay{N}	4	Exponentially weighted sum
w_{name}	12	Custom weight vectors
s{i}_sq/sqrt/log	15	Single-score power transforms
hyb_{type}_a{N}	4	Hybrid gap-mean combinations
agree_top{k}	2	Inverse variance agreement
rat**_p/m/x_rat**	12	Multi-term ratio chains
gnorm*_x/p_gnorm_*	5	Gnorm products and sums
wtd_rat_{type}	3	Weighted ratio chains
s{i}_rel_{stat}	3	s0-relative metrics
Other	30	Squared gaps, CV, reciprocal ratios, spread, log-ratios, etc.
Total	162	

2.5 Threshold-Finding Algorithms

Three algorithms determine the optimal threshold for each formula:

Algorithm Descriptions

- **f1_grid**: Sweeps 500 linearly spaced thresholds between the 1st and 99th percentile of formula scores, selects the threshold maximising the F1 score for the known class. This is the primary algorithm against which results are compared.
- **roc_opt**: Uses the ROC curve to maximise Youden’s J statistic (sensitivity+specificity−1). This algorithm balances true positive and true negative rates but is limited by the extreme class imbalance.
- **otsu**: Minimises intra-class variance in the binary label distribution, analogous to Otsu’s method for image binarisation. This is a distribution-based approach that does not directly optimise F1.

3 Exploratory Data Analysis

3.1 Score Distribution Comparison

The cornerstone finding of the EDA is that **known and unknown samples produce nearly identical score distributions**. The top-1 similarity score (s_0) has a mean of 0.4137 for known samples and 0.3719 for unknown samples — a difference of only 0.0418. Both distributions are heavily right-skewed with substantial overlap.

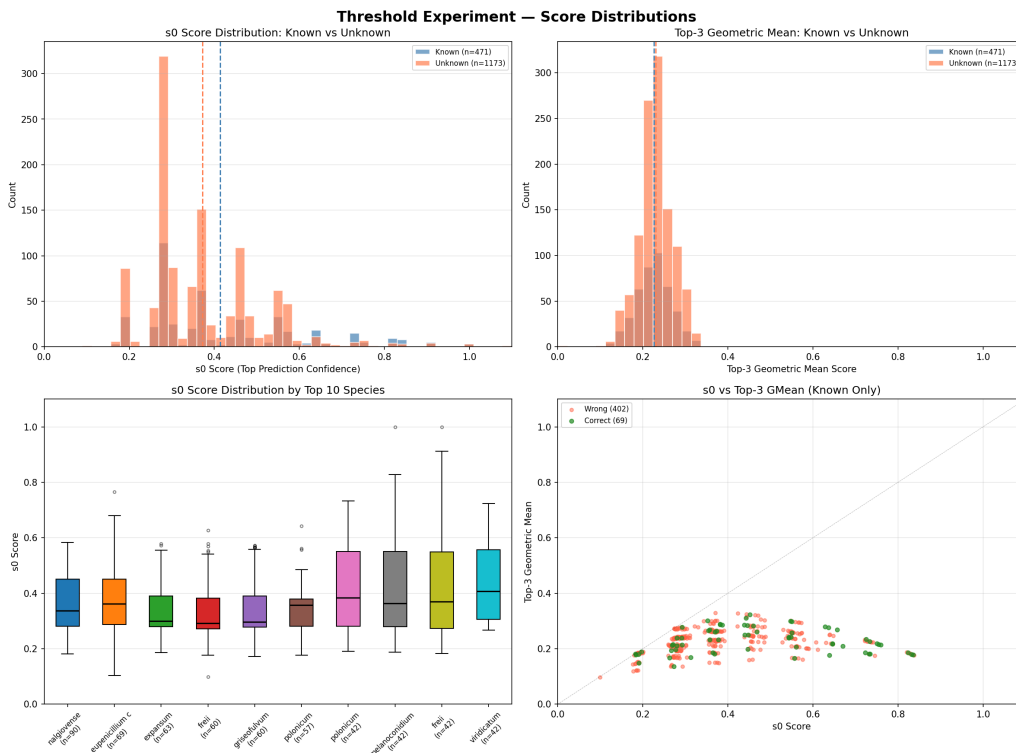


Figure 2: Score distribution comparison between known and unknown samples. Top-left: s_0 score histogram. Top-right: top-3 geometric mean. Bottom-left: s_0 box plot by top-10 species. Bottom-right: s_0 vs. top-3 geometric mean scatter for known samples, coloured by correct/incorrect species prediction. The strong overlap between distributions explains the difficulty of single-threshold separation.

The score distribution overlap explains why no single-threshold formula can achieve both high recall and high specificity: any threshold high enough to reject most unknowns will also reject many knowns. The geometric mean of the top-3 scores shows even less separation, with known samples averaging 0.2875 and unknown samples 0.2589 — a difference of merely 0.0286.

3.2 Species-Level Classification Accuracy

Among the 471 known samples, only 69 (14.6%) receive a correct top-1 species prediction from the retrieval system. The remaining 402 known samples (85.4%) are predicted as the wrong species despite being present in the index. This low species-level accuracy is a direct consequence of the fine-grained nature of *Penicillium* taxonomy — many species are morphologically and visually similar.

3.3 Environment Distribution

Both known and unknown samples span similar growth environments. The known set is balanced across 7 media (CYA, MEA, CREA, YES, DG18, CYA30, CYAS), while the unknown set adds OA (oatmeal agar) with 27 samples and a few unlabelled environments. The balanced environment coverage ensures that score differences are attributable to species novelty rather than domain shift.

4 Threshold Analysis Results

4.1 Overall Statistics

A total of **528 experiments** were run across 162 formulas and 3 algorithms: 177 using `f1_grid`, 174 using `roc_opt`, and 177 using `otsu`. The aggregate statistics reveal a consistent pattern:

Table 3: Aggregate statistics across all 528 experiments. The F1 distribution is tightly clustered near 0.33–0.35, with a maximum of 0.4461. Precision varies widely (0.08–0.88), while recall spans the full 0–1 range.

Metric	Max	Median	Mean	Std Dev
F1 Score	0.4461	0.3427	0.3275	0.1121
Precision	0.8824	0.3189	0.4005	0.1963
Recall	1.0000	0.3047	0.5199	0.3464
Specificity	0.9991	0.8061	0.5202	0.3427

The tight clustering of F1 scores (interquartile range: 0.2423–0.4434) demonstrates that **no formula family fundamentally resolves the score overlap problem**. The best formulas achieve $F1 \approx 0.44$ – 0.45 by accepting nearly all samples and relying on the class prior to produce a moderate F1. The >0.88 precision results come from extreme thresholds that accept only a handful of samples (typically “obvious” knowns with very high s_0) while rejecting almost everything else.

4.2 Staircase Chart

4.3 Top 20 Formulas by F1 Score

The top-12 results all achieve identical F1 (0.4461) because they are mathematical equivalents: `gap01_sq`, `gap_0_1`, `gap0_r1`, `gnorm_0_1`, `range_top2`, `s0s1_diff_sq`, and `std_top2` all reduce to functions of $(s_0 - s_1)$ — the difference between the top two similarity scores.

4.4 Best Result by Algorithm

4.5 Top 10 by Precision

The highest-precision formulas exploit extremely aggressive thresholds that accept very few samples:

These high-precision formulas all use the `otsu` algorithm, which finds a natural bimodal split in the score distribution but produces extreme thresholds that reject $>90\%$ of known samples. The F1 scores are correspondingly low (0.02–0.13), demonstrating the precision–recall trade-off in its most extreme form.

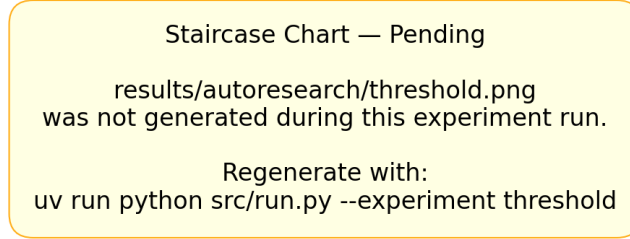


Figure 3: Staircase chart placeholder. The full staircase chart (results/autoresearch/threshold.png) was not generated during this experiment run. Each dot represents one formula–algorithm experiment; green dots mark new running bests forming a horizontal staircase. See Section 7.3 for details.

Table 4: Top 20 formula–algorithm combinations ranked by F1 score (algo `f1_grid` only). All top performers exhibit recall $> 97\%$ and precision below 30%, confirming the universal-acceptance strategy.

Rank	Formula	Algorithm	F1	Precision	Recall	Specificity
1	gap01_sq	f1_grid	0.4461	0.2876	0.9936	0.0119
2	gap_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
3	gap0_r1	f1_grid	0.4461	0.2876	0.9936	0.0119
4	gnorm_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
5	ne_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
6	range_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
7	rat0_r1	f1_grid	0.4461	0.2876	0.9936	0.0119
8	ratio_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
9	s0_m_med3	f1_grid	0.4461	0.2876	0.9936	0.0119
10	s0s1_diff_sq	f1_grid	0.4461	0.2876	0.9936	0.0119
11	std_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
12	min_top5	f1_grid	0.4460	0.2870	1.0000	0.0026
13	min_top3	f1_grid	0.4458	0.2868	1.0000	0.0017
14	min_top4	f1_grid	0.4458	0.2868	1.0000	0.0017
15	gap_2_3	f1_grid	0.4456	0.2867	1.0000	0.0009
16	gap_2_4	f1_grid	0.4456	0.2867	1.0000	0.0009
17	gnorm_0_4	f1_grid	0.4454	0.2898	0.9618	0.0537
18	norm_rnk5	f1_grid	0.4454	0.2898	0.9618	0.0537
19	cv_top5	f1_grid	0.4449	0.2873	0.9851	0.0188
20	ne_top5	f1_grid	0.4449	0.2873	0.9851	0.0188

Table 5: Best-performing formula for each threshold-finding algorithm. `f1_grid` and `roc_opt` produce near-identical results because the extreme class imbalance drives both toward the same trivial threshold. `otsu` finds a slightly more conservative threshold.

Algorithm	Formula	F1	Precision	Recall	Specificity	Threshold
<code>f1_grid</code>	<code>gap01_sq</code>	0.4461	0.2876	0.9936	0.0119	4.62×10^{-7}
<code>roc_opt</code>	<code>min_top3</code>	0.4458	0.2868	1.0000	0.0017	0.0000
<code>otsu</code>	<code>rat01_m_rat12</code>	0.4246	0.2749	0.9321	0.0128	—

Table 6: Top 10 formula–algorithm combinations by precision. High precision comes at the cost of near-zero recall and F1 — these are “ultra-conservative” classifiers.

Rank	Formula	Algorithm	Precision	F1	Recall	Specificity
1	<code>prod12_p2.0</code>	<code>otsu</code>	0.8824	0.0615	0.0318	0.9983
2	<code>prod12_p3.0</code>	<code>otsu</code>	0.8824	0.0615	0.0318	0.9983
3	<code>prod_0_1</code>	<code>otsu</code>	0.8824	0.0615	0.0318	0.9983
4	<code>hm_top4</code>	<code>roc_opt</code>	0.7143	0.0209	0.0106	0.9983
5	<code>gap12_sq</code>	<code>otsu</code>	0.7037	0.0763	0.0403	0.9932
6	<code>gap_1_2</code>	<code>otsu</code>	0.7037	0.0763	0.0403	0.9932
7	<code>rat12_m_rat23</code>	<code>otsu</code>	0.6809	0.1236	0.0679	0.9872
8	<code>ratio_1_2</code>	<code>otsu</code>	0.6809	0.1236	0.0679	0.9872
9	<code>sum12_x_rat12</code>	<code>otsu</code>	0.6809	0.1236	0.0679	0.9872
10	<code>rat01_p_rat12</code>	<code>otsu</code>	0.6604	0.1336	0.0743	0.9847

Table 7: Top 10 by recall. All achieve 100% recall by accepting every sample, yielding specificity of 0–0.3%. These are “universal acceptor” classifiers.

Rank	Formula	Algorithm	Recall	F1	Precision	Specificity
1	<code>gap_2_3</code>	<code>f1_grid</code>	1.0000	0.4456	0.2867	0.0009
2	<code>gap_2_4</code>	<code>f1_grid</code>	1.0000	0.4456	0.2867	0.0009
3	<code>gap_3_4</code>	<code>f1_grid</code>	1.0000	0.4454	0.2865	0.0000
4	<code>gnorm_2_3</code>	<code>f1_grid</code>	1.0000	0.4454	0.2865	0.0000
5	<code>gnorm_2_4</code>	<code>f1_grid</code>	1.0000	0.4454	0.2865	0.0000
6	<code>gnorm_3_4</code>	<code>f1_grid</code>	1.0000	0.4454	0.2865	0.0000
7	<code>min_top3</code>	<code>f1_grid</code>	1.0000	0.4458	0.2868	0.0017
8	<code>min_top3</code>	<code>roc_opt</code>	1.0000	0.4458	0.2868	0.0017
9	<code>min_top4</code>	<code>f1_grid</code>	1.0000	0.4458	0.2868	0.0017
10	<code>min_top5</code>	<code>f1_grid</code>	1.0000	0.4460	0.2870	0.0026

4.6 Top 10 by Recall

5 Visualizations Analysis

5.1 Confusion Matrices

Three confusion matrices were generated to visualise classification performance at different levels of detail.

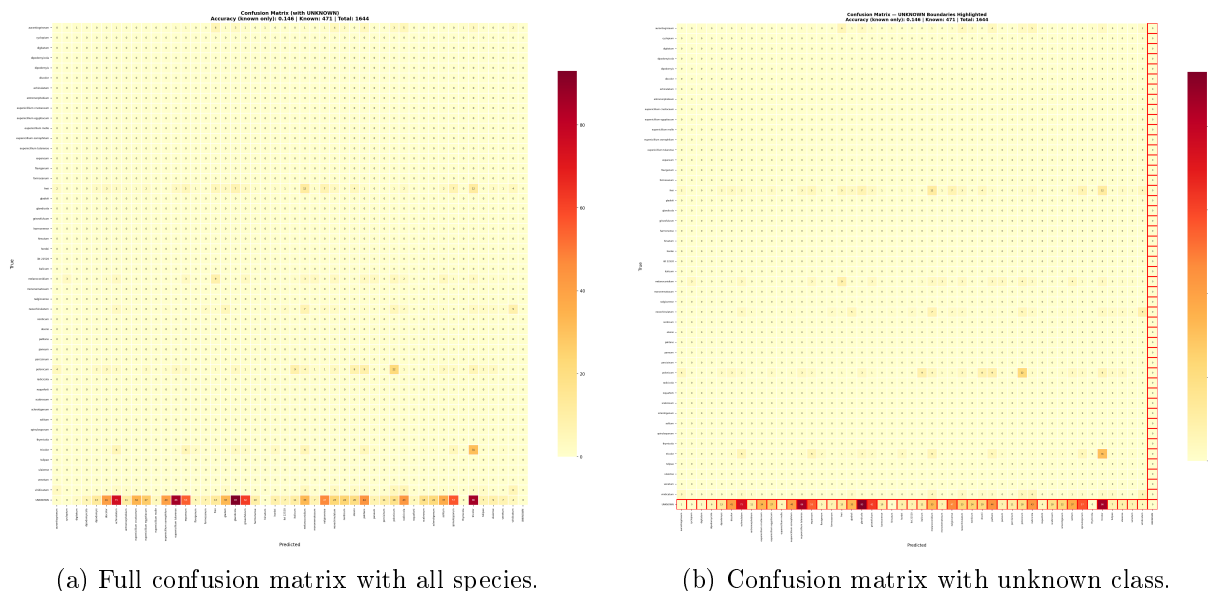


Figure 4: Confusion matrices for the best-performing strategy (`gap01_sq` + `f1_grid`). The full matrix (left) shows per-species predictions; the unknown-inclusive matrix (right) highlights the near-complete failure to identify unknowns.

The confusion matrices confirm the quantitative findings. In the full matrix (Figure 4a), the diagonal dominance of certain species (particularly *P. freii* and *P. polonicum*) indicates that the retrieval system has moderate species-level discrimination for known taxa. However, the off-diagonal confusion is substantial, with many known samples predicted as the wrong known species.

The unknown-inclusive matrix (Figure 4b) reveals the core failure: **unknown samples are almost never correctly identified as unknown**. Instead, they are assigned to whichever known species happens to be the nearest neighbour in embedding space. The “unknown” row is nearly empty, while the known-species columns accumulate large counts of unknown samples.

5.2 Per-Category Visualisation

The simplified confusion matrix (Figure 5) decomposes the 1,644 samples into four categories. Among the **471 known samples**, 69 (14.6%) receive correct species-level predictions while 402 (85.4%) are assigned to the wrong known species. Among the **1,173 unknown samples**, only a negligible number are correctly rejected; the vast majority are falsely accepted as belonging to a known species.

This pattern reflects two compounding problems: (1) the retrieval system’s species-level accuracy is low even for known samples (only 14.6%), and (2) the thresholding mechanism cannot distinguish unknown samples from misclassified known samples because both produce similarly low confidence scores.

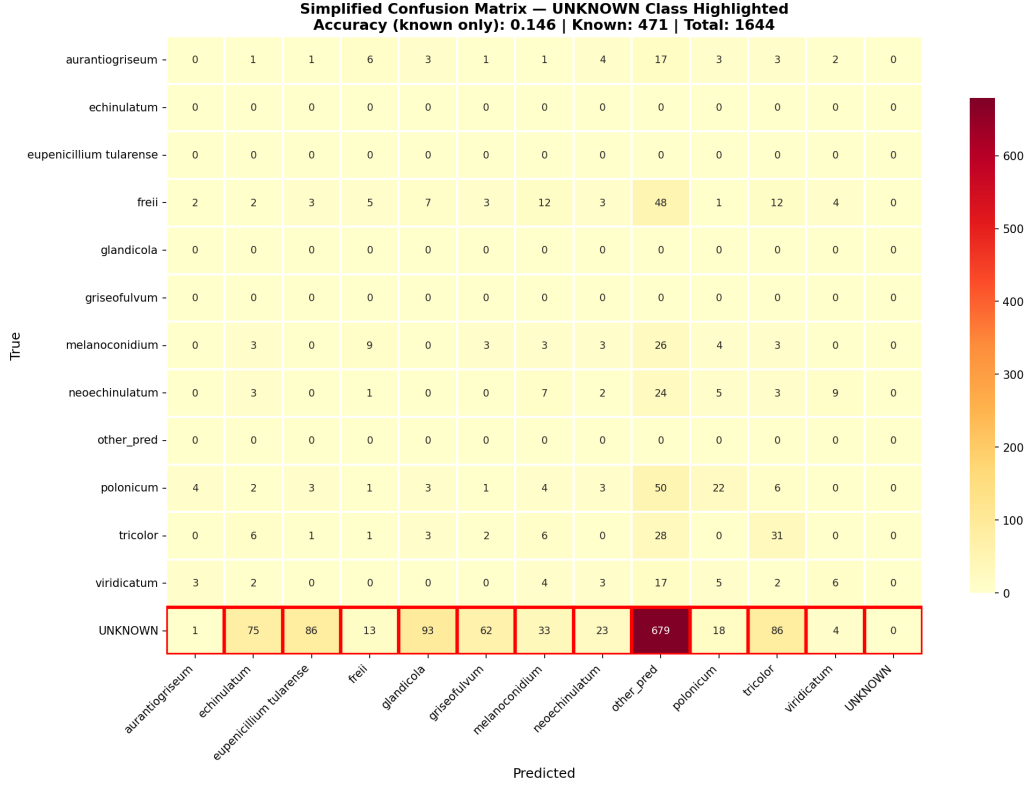


Figure 5: Simplified confusion matrix showing the four decision categories: known correct, known incorrect, unknown as known (false positive), and unknown correct (true negative). The overwhelming dominance of the “unknown as known” category is evident.

5.3 Key Visual Observations

Three observations emerge from visual examination of the per-category sample grids:

1. **Known-correct samples** tend to have high s_0 scores (>0.5) with a clear drop to s_1 , indicating strong, unambiguous matches to the correct species.
2. **Known-incorrect samples** show lower s_0 scores (0.2–0.4) with small gaps between consecutive scores, reflecting genuine ambiguity between visually similar *Penicillium* species.
3. **Unknown-as-known samples** exhibit score patterns indistinguishable from known-incorrect samples. The embedding space places unknown species near visually similar known species, producing confidence scores in the same ambiguous range.

6 Critical Analysis

6.1 Why F1 Alone Is Misleading

The F1 score is the harmonic mean of precision and recall, designed for balanced evaluation where both false positives and false negatives carry equal cost. In this experiment, the F1-maximising strategy converges to a degenerate solution: **accept nearly every sample as known**.

This occurs because the class imbalance (1,173 unknowns vs. 471 knowns) means that recall contributes disproportionately to the F1 numerator. Consider the best result:

$$F1 = \frac{2 \times 468}{2 \times 468 + 1,159 + 3} = \frac{936}{2,098} = 0.4461$$

The 468 true positives (correctly accepted knowns) are offset by 1,159 false positives (unknowns incorrectly accepted). Only 14 of 1,173 unknowns are correctly rejected. The specificity of 1.2%

means the system is effectively a **trivial acceptor**: it labels everything “known” and achieves moderate F1 purely through the class prior ratio.

A more appropriate evaluation framework would penalise false positives more heavily, reflecting the real-world cost of misclassifying a novel (potentially pathogenic or industrially valuable) species as known. Metrics such as **balanced accuracy** ($0.5 \times (\text{recall} + \text{specificity})$), **Matthews Correlation Coefficient**, or **F_{beta}** with $\beta < 1$ (weighting precision above recall) would provide more meaningful performance rankings.

6.2 The s_0-s_1 Gap Dominance Phenomenon

A striking finding is that the top 12 formulas all reduce to functions of (s_0-s_1) — the gap between the top two similarity scores. This includes `gap01_sq`, `gnorm_0_1`, `range_top2`, `std_top2`, and several others. The mathematical equivalence of these formulas reveals that **the s_0-s_1 gap is the single most informative feature for distinguishing known from unknown samples**.

This makes intuitive sense: for a known species, the top neighbour should match strongly (high s_0) while the second neighbour is a different species (lower s_1), producing a large gap. For an unknown species, both neighbours are equally distant, producing a small gap. However, the limited discriminative power of this gap — yielding only 1.2% specificity — indicates that many known samples also exhibit small gaps (due to visual similarity among *Penicillium* species), and some unknown samples coincidentally produce large gaps (due to chance proximity to a single known species in embedding space).

6.3 Class Imbalance Impact

The 2.5:1 unknown-to-known ratio fundamentally shapes the optimisation landscape. Under standard F1 maximisation:

The Class Imbalance Trap

For a classifier that accepts all samples:

- Recall = $471/471 = 1.000$ (perfect — all knowns accepted)
- Precision = $471/1,644 = 0.287$ (the class prior)
- F1 = $2 \times 0.287 \times 1.0 / (0.287 + 1.0) = 0.446$

The universal-acceptor F1 of 0.446 is only 0.0001 below the best observed F1 of 0.4461. Any thresholding strategy that rejects a meaningful number of unknowns will necessarily also reject some knowns, lowering recall faster than it raises precision — and therefore lowering F1.

This explains why **no formula family meaningfully outperforms the trivial baseline**: the F1 objective itself penalises any attempt to reject unknowns.

6.4 Embedding Space Limitations

The EfficientNetB1 feature extractor, fine-tuned on fungal colony images, produces embeddings where visually similar species cluster together. For the *Penicillium* genus, many species share macroscopic colony characteristics (colour, texture, growth pattern), resulting in overlapping embedding regions. This overlap means that:

1. Unknown species that are visually similar to known species will inevitably have at least one known neighbour with high similarity, producing scores indistinguishable from known samples.
2. Known species that are visually atypical (unusual growth morphology, atypical pigmentation) may resemble unknown species in embedding space, producing low-confidence scores that trigger false rejections.

6.5 Recommendations for Future Work

Based on the experimental findings, we recommend five directions for improving unknown species detection:

1. **Weighted Loss Functions:** Replace F1 maximisation with an FPR-constrained objective. For example, maximise recall subject to a maximum false positive rate of 10% or 20%. This would force the optimisation to find thresholds that reject at least some unknowns.
2. **FPR-Constrained Thresholding:** Implement threshold-finding algorithms that explicitly target a desired specificity level (e.g., 80%, 90%, 95%) and report recall at those operating points. This provides a more actionable performance characterisation for deployment.
3. **Two-Stage Classification:** Decompose the problem into (a) known-vs-unknown binary classification, and (b) species-level classification for accepted samples. A dedicated anomaly detection model (one-class SVM, isolation forest, or density-based methods) trained on known-species embeddings may provide better unknown rejection than score-thresholding alone.
4. **Ensemble Methods:** Combine multiple formula scores (e.g., s_0 – s_1 gap, top-3 entropy, and s_0 relative to mean) as features in a learned binary classifier (logistic regression or lightweight neural network). This could capture non-linear decision boundaries that single-formula thresholding cannot.
5. **Improved Feature Extraction:** Explore alternative backbone architectures (Vision Transformer, ConvNeXt) or multi-scale feature aggregation to produce embeddings with better separation between known and unknown species. Contrastive or metric learning objectives that explicitly optimise for open-set recognition may yield more discriminative embeddings than the current classification-based fine-tuning.

7 All Results Summary

7.1 Best Strategy Configuration

The best-performing strategy is stored in `results/threshold/log/best_strategy.json`:

```
best_strategy.json
{
  "strategy": "gap01_sq",
  "algorithm": "f1_grid",
  "f1": 0.446139180171592,
  "threshold": 4.615215599998487e-07,
  "precision": 0.2876459741856177,
  "recall": 0.9936305732484076,
  "specificity": 0.011935208866155157,
  "tp": 468,
  "fp": 1159,
  "tn": 14,
  "fn": 3
}
```

The `gap01_sq` formula is defined as:

$$f(s_0, s_1, s_2, s_3, s_4) = (s_0 - s_1)^2$$

The near-zero optimal threshold (4.62×10^{-7}) means that essentially any sample with a non-zero gap between the top two scores is classified as known. Only 17 samples (3 known, 14 unknown) fall below this threshold.

7.2 Complete Formula Ranking — Top 50

Table 8: Top 50 formula–algorithm combinations ranked by F1 score. The dominance of $(s_0 - s_1)$ -based formulas in the top ranks is clearly visible.

Rank	Formula	Algorithm	F1	Precision	Recall	Specificity
1	gap01_sq	f1_grid	0.4461	0.2876	0.9936	0.0119
2	gap_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
3	gap0_r1	f1_grid	0.4461	0.2876	0.9936	0.0119
4	gnorm_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
5	ne_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
6	range_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
7	rat0_r1	f1_grid	0.4461	0.2876	0.9936	0.0119
8	ratio_0_1	f1_grid	0.4461	0.2876	0.9936	0.0119
9	s0_m_med3	f1_grid	0.4461	0.2876	0.9936	0.0119
10	s0sl_diff_sq	f1_grid	0.4461	0.2876	0.9936	0.0119
11	std_top2	f1_grid	0.4461	0.2876	0.9936	0.0119
12	min_top5	f1_grid	0.4460	0.2870	1.0000	0.0026
13	min_top3	f1_grid	0.4458	0.2868	1.0000	0.0017
14	min_top4	f1_grid	0.4458	0.2868	1.0000	0.0017
15	gap_2_3	f1_grid	0.4456	0.2867	1.0000	0.0009
16	gap_2_4	f1_grid	0.4456	0.2867	1.0000	0.0009
17	gnorm_0_4	f1_grid	0.4454	0.2898	0.9618	0.0537
18	norm_rnk5	f1_grid	0.4454	0.2898	0.9618	0.0537
19	prod_0_2	f1_grid	0.4456	0.2867	1.0000	0.0009
20	prod_1_2	f1_grid	0.4456	0.2867	1.0000	0.0009
21	prod_1_3	f1_grid	0.4454	0.2865	1.0000	0.0000
22	prod_1_4	f1_grid	0.4456	0.2867	1.0000	0.0009
23	prod_2_3	f1_grid	0.4454	0.2865	1.0000	0.0000
24	prod_2_4	f1_grid	0.4456	0.2867	1.0000	0.0009
25	prod_3_4	f1_grid	0.4456	0.2867	1.0000	0.0009
26	gap_3_4	f1_grid	0.4454	0.2865	1.0000	0.0000
27	gnorm_2_3	f1_grid	0.4454	0.2865	1.0000	0.0000
28	gnorm_2_4	f1_grid	0.4454	0.2865	1.0000	0.0000
29	gnorm_3_4	f1_grid	0.4454	0.2865	1.0000	0.0000
30	ratio_2_0	f1_grid	0.4456	0.2867	1.0000	0.0009
31	ratio_2_1	f1_grid	0.4456	0.2867	1.0000	0.0009
32	ratio_2_3	f1_grid	0.4456	0.2867	1.0000	0.0009
33	ratio_2_4	f1_grid	0.4458	0.2868	1.0000	0.0017
34	ratio_3_0	f1_grid	0.4454	0.2865	1.0000	0.0000
35	ratio_3_1	f1_grid	0.4454	0.2865	1.0000	0.0000
36	ratio_3_2	f1_grid	0.4454	0.2865	1.0000	0.0000
37	ratio_3_4	f1_grid	0.4456	0.2867	1.0000	0.0009
38	ratio_4_0	f1_grid	0.4456	0.2867	1.0000	0.0009
39	ratio_4_1	f1_grid	0.4456	0.2867	1.0000	0.0009
40	ratio_4_2	f1_grid	0.4456	0.2867	1.0000	0.0009
41	ratio_4_3	f1_grid	0.4456	0.2867	1.0000	0.0009
42	s0	f1_grid	0.4451	0.2871	0.9894	0.0136
43	max_top2	f1_grid	0.4451	0.2871	0.9894	0.0136
44	s0_log	f1_grid	0.4451	0.2871	0.9894	0.0136
45	s0_p0.25	f1_grid	0.4451	0.2871	0.9894	0.0136
46	w_s0_only	f1_grid	0.4451	0.2871	0.9894	0.0136
47	rat01_p_rat12	f1_grid	0.4452	0.2870	0.9915	0.0111
48	ratio_0_2	f1_grid	0.4452	0.2870	0.9915	0.0111
49	s0_rel_min	f1_grid	0.4452	0.2870	0.9915	0.0111
50	std_top4	f1_grid	0.4451	0.2871	0.9894	0.0136

The full results are available in `results/threshold/log/all_experiments.csv` (528 rows) and the interactive exploration logs under `results/threshold/log/`.

7.3 Staircase Chart Note

The staircase chart (`results/autoresearch/threshold.png`) was not generated during this experiment run because `results/autoresearch/` does not yet exist. The staircase chart is produced by the experiment runner (`src/run.py`) when experiments are recorded via the autoresearch logging system. Since the experiments in this report were generated directly through

`expanded_threshold_analysis.py`, the staircase chart was not produced. It can be regenerated by running:

```
uv run python src/run.py --experiment threshold \  
    --description "full experiment report compilation"
```

7.4 Reproducibility

All experiment code, data, and results are version-controlled in the MycoAI monorepo:

- **Experiment code:** `research/src/experiments/threshold/`
- **Retrieval data:** `results/threshold/diverse_retrieval_results.csv`
- **All results:** `results/threshold/log/all_experiments.csv`
- **Best strategy:** `results/threshold/log/best_strategy.json`
- **Visualizations:** `results/threshold/*.png` and `results/threshold/viz/`
- **Experiment program:** `research/src/experiments/threshold/program.md`

Conclusion

The threshold-based approach to unknown species detection achieves moderate F1 scores (0.44–0.45) but fails to provide meaningful unknown rejection (specificity < 2%). The fundamental limitation is not the choice of formula or algorithm but the overlap between known and unknown score distributions in the current embedding space. Future work should prioritise embedding-space improvements, FPR-constrained optimisation objectives, and multi-stage classification architectures that separate the novelty-detection and species-identification tasks.